THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of chromosomes in a human sperm is 23 and in the egg is 23. In a normal human zygote and a skin cell, the numbers are respectively 1

(a) 46 and 46 **(b)** 23 and 46 **(c)** 23 and 23 **(d)** 46 and 23

2. Assertion (A): Variations are important for the survival of species. Reason (R): They provide a chance to adapt to changing conditions. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The probability that a couple's four children are all girls is 1

(a) 1/2 **(b)** 1/4 **(c)** 1/8 **(d)** 1/16

4. A population of asexually reproducing bacteria shows less variation than a population of humans because 1

(a) bacteria have no DNA
(b) sexual reproduction reduces variation
(c) humans have no mutation
(d) there is only one parent and only copying errors

5. Assertion (A): Children resemble their parents but are not identical. Reason (R): They inherit genes from both parents in new combinations. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Of the children of a couple, which parent's chromosome decides that a son carries a Y? 1

(a) Mother **(b)** Father **(c)** Both **(d)** Neither

7. A plant with TtRr genotype is self-pollinated. The probability of an offspring being dwarf and wrinkled is 1

(a) 1/4 **(b)** 1/16 **(c)** 3/16 **(d)** 9/16

8. Assertion (A): A test cross is carried out with a recessive parent. Reason (R): Recessive parent produces only one type of gamete. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A father passes the same sex chromosome to all his sons. It is the 1

(a) Y chromosome **(b)** X chromosome **(c)** both X and Y **(d)** none

10. The gene for a protein that makes a plant growth hormone is defective. The plant will likely be 1

(a) dwarf **(b)** unchanged **(c)** tall **(d)** colourless

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If both parents are TT and tt, what is the probability that the F2 is tt? Explain. 2

12. Find the probability that a child is a girl if the father's sperm carries a Y chromosome. 2

13. How many types of gametes will be produced by TtRrYy? Show the reasoning. 2

14. Why can't an individual with recessive phenotype be heterozygous? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the fallacy in the statement: A man who exercises regularly will have sons with strong muscles. 3

16. A tall plant (Tt) is self-pollinated and gives 4 offspring. Can we say one will be dwarf? Explain. 3

17. What is the effect of the environment on a person's height even if the genes for height are the same? 3

OR

Why is the chance of the first child being a boy the same as the chance of the second child being a boy? 3

18. Study the statement: Mendel's F1 had no dwarf plants, so the dwarf trait was lost. Evaluate. 3

OR

A sperm contains 22 autosomes and X. Name the sex of the child and explain how the child gets its other sex chromosome. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. In a dihybrid cross of RrYy x RrYy, find 5

(a) the fraction of seeds that are round yellow,

(b) the fraction that are homozygous for both genes, and

(c) the fraction that are both heterozygous.

OR

Describe how the knowledge of heredity is used to explain why an organism that has the same DNA as its clone-mate can look different. 5

20. Explain how Mendel's experiments and the idea of genes on chromosomes together explain why the F2 of a dihybrid cross has a 9 : 3 : 3 : 1 ratio. 5

OR

In humans, Rh positive is dominant over Rh negative. A heterozygous woman marries a Rh negative man. Show the possible children with their genotypes, phenotypes and the probability of each. 5

END OF QUESTION PAPER Total: 40 Marks